

3(a)(i)	The first law of thermodynamics states that the <u>increase in the internal energy ΔU of a system is equal to the sum of the heat Q supplied to the system and work done W on the system.</u> The <u>latent heat of vaporisation is considerably higher</u> because evaporation involve a considerably <u>greater molecular separation and volume increase than melting.</u> Hence, it involves a considerably greater increase in potential energy, and considerably <u>more work has to be done by the system.</u>	B1 B1 B1 B1
3(a)(ii)	Mass of the ethanol = $0.79 \times 2 = 1.58 \text{ g}$ Energy required from 30°C to $78^\circ\text{C} = 2.4 \times 1.58 \times (78 - 30) = 182 \text{ J}$ Energy required for boiling = $840 \times 1.58 = 1327 \text{ J}$ Required thermal energy = $1327 + 182 = 1509 \text{ J}$	C1 C1 C1 A1
3(b)(i)	$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$ $\frac{2.6 \times 10^6 \times 2.9 \times 10^{-5}}{790} = \frac{P_2 \times 2.9 \times 10^{-4}}{314}$ Pressure of the gas = 103000 Pa	M1 A1
3(b)(ii)	$\Delta U = Q + W$ $= 0 + -91$ $= -91 \text{ J}$ Change in internal energy = -91 J	A1
3(b)(iii)	Internal energy of a system is the <u>sum of kinetic energies and potential energies</u> associated with the gas molecules. As the gas <u>behaves ideally, internal energy is just the sum of kinetic energies</u> of the gas molecules and it is <u>proportional to the temperature</u> of the gas system. A decrease in the internal energy of the gas that behaved ideally indicates a decrease in the average kinetic energy of the molecules, which would be reflected as a drop in temperature.	B1 B1